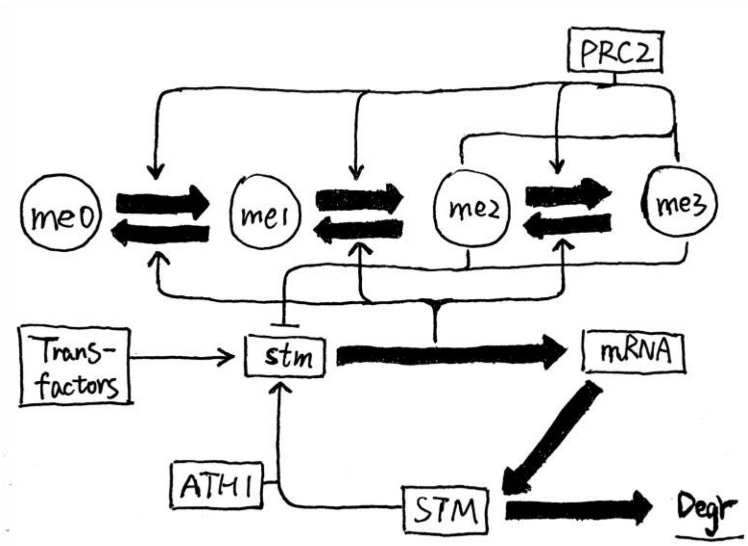
Computational Methods and Simulation Details

Mathematical Model



Overview

We constructed a biological process model of the *SHOOT MERISTEMLESS* (*STM*) gene locus to explain our experimental results by expanding a previous mathematical model of a generic polycomb target gene[1,2]. The new model is composed of three main inter-coupled sections, i.e., chromatin histone modification, *STM* expression cycle and DNA replication/cell division. The dynamics of H3K27 methylation, demethylation, gene transcription and translation, protein degradation, as well as DNA replication were simulated using the Gillespie's stochastic simulation algorithm (SSA)[3], a classical and widely used method for simulating the stochastic behavior of chemical systems against time evolution by Monte Carlo sampling. This algorithm is defined by a series of possible reactions or events, and a corresponding propensity for each event to occur (master equation). At each iteration, the time interval Δt for the next coming event and the event itself are both selected probabilistically based on Poisson process. Subsequently, the system is updated by performing the selected event, and time is incremented by Δt.

In our simulations, we tracked the methylation status at H3K27, denoted by $H_i\ (i = 1,2,\dots,N_\mathrm{h})$, of each H3 histone within the *STM* locus on chromatin, where $H_i \in \{\mathrm{me0}, \mathrm{me1}, \mathrm{me2}, \mathrm{me3}\}$ represents the number of methyl group. Here we set $N_\mathrm{h} = 36$ to approximate the physical length of *STM* gene (~3482 bp) and its promoter, according to the putative value that a nucleosome contains ~200 bp (include linkers). Histones with the indices $2j-1$ and $2j$, for each positive integer j, were thought to be in the same nucleosome. Meanwhile, we recorded the number of STM protein molecules (actually we did not restrict its values to integers, discuss later), denoted by N_p, in each time step to characterize the *STM* expression level and stem cell fate. Overall, we considered totally $2N_\mathrm{h}+2$ possible events

(N_h histone methylations, N_h histone demethylations, gene expression and protein degradation) in our model. Note that, not all events are possible at any time, e.g., methylation of trimethylated histones or degradation when proteins are no longer presented. The event propensities, which are elaborated in the following text, denoted by r with specific superscripts, are recalculated after each system update. The summary of model parameters is shown in Table S1.

Chromatin Histone Modification

H3K27 methylation. In our model, polycomb repressive complex 2 (PRC2) activity results in the methylation of H3K27. PRC2 is a large protein complex containing an enzymatic subunit with histone methyltransferase activity[4-7], and a non-catalytic subunit that recognizes H3K27me3/me2[8]. Therefore, PRC2 can recognize existing H3K27me3/me2 and then add methyl group to other histones via a read/write mechanism[8-11], forming a conserved mechanism in both animals and plants. Meanwhile, both *in vitro* and *in vivo* experimental evidence suggested that PRC2 mostly worked in a non-processive manner[12,13], adding one methyl group at a time, which we applied in our simulation.

To model H3K27 methylation, we followed the ideas of a previous work[1], but with slight differences. Overall, the propensity of methylation, r_i^{me} for histone i, depends on the current methylation status of the surrounding histones, together with some unusual long-range interactions through DNA looping. Besides, r_i^{me} also depends on two reference rates: the PRC2-dependent methylation rate k_{me} and stochastic methylation rate γ_{me}. Moreover, our model also takes the different catalytic abilities of PRC2 on specific methylated substrates (H3K27me0/me1/me2) into account, as supported by *in vitro* experiments[12]. For each i, the methylation reaction propensity is calculated as

$$\begin{aligned} r_i^{\mathrm{me}} = \beta[&\delta_{H_i,\mathrm{me0}}(\gamma_{\mathrm{me01}} + k_{\mathrm{me01}}E_i) + \\ &\delta_{H_i,\mathrm{me1}}(\gamma_{\mathrm{me12}} + k_{\mathrm{me12}}E_i) + \\ &\delta_{H_i,\mathrm{me2}}(\gamma_{\mathrm{me23}} + k_{\mathrm{me23}}E_i)] \end{aligned} \qquad \text{(Equation S1)}$$

where $\delta_{x,y}$ is the Kronecker delta symbol that equals 1 if $x = y$ else 0.

To incorporate the positive feedback between PRC2 and H3K27me3/me2 (but not H3K27me1), the model allows these two repressive marks to activate PRC2 and promote methylation on neighboring histones. Therefore, we defined a dimensionless enhancement factor for histone i, denoted as E_i in Equation S1, to capture this effect

$$E_i = \sum_{j \in M_i \cap [1,\ldots,N_\mathrm{h}]} \left(\rho\delta_{H_j,\mathrm{me2}} + \delta_{H_j,\mathrm{me3}}\right) + e_{\mathrm{distal}} \qquad \text{(Equation S2)}$$

where M_i, called the neighboring set, is composed of histones that lie in the same and adjacent nucleosomes

$$M_i = \begin{cases} \{i-3, i-2, i-1, i+1, i+2\}, \text{ if } i \text{ is even} \\ \{i-2, i-1, i+1, i+2, i+3\}, \text{ if } i \text{ is odd} \end{cases} \qquad \text{(Equation S3)}$$

In our model, the first term in Equation S2 accounts for the proximal gain, and ρ represents the reduced efficiency (~10-fold) of activated PRC2 by H3K27me2 relative to H3K27me3[8]. The second term, e_{distal} in Equation S2, captures the contribution of long-range interactions, or distal gain, when DNA looping brings together nucleosomes that are far from each other in one dimension[11,14], although this event is less frequent than the others. Note that, in Equation S2, histones outside the *STM* locus are ignored. As a result, this introduces a slight bias toward the active state, especially of histones on boundary nucleosomes, as the boundary histones only undergo one-sided recruitment. However, this bias is expected to be inconsequential.

The β in Equation S1 reflects the effective activity of PRC2. In this study, we identified that PRC2 can be recruited to *STM* locus by two classes of transcription factor, i.e., *RABBIT EARS* (*RBE*) and *BASIC PENTACYSTEINE* (*BPC*), in a sequence-specific manner. Since the pre-procedures of methylation (e.g., recruitment, migration, binding, allostery, catalysis etc.) all take some times[15], in a cell with extremely rapid division frequency, the PRC2 may not even have enough time to finish work at all before the cell enters a new cell cycle. Hence, we expect that $\beta \to 0$ when cell cycle $T \to 0$. Meanwhile, a study proved that PRC2 has more sufficient time to complete methylation for H3K27 when G1-phase is extended, leading to a significant increase in H3K27me3 level[16]. Finally, we introduced a scaled cumulative distribution function (CDF) of exponential distribution to incorporate the above effect on β

$$\beta = \beta(T) = \beta_{\max}\left(1 - \exp\left(-\frac{bT}{T_0}\right)\right) \qquad \text{(Equation S4)}$$

where T_0, b and $\beta_{\max}$ is the normalized cell cycle, exponential factor (positive) and the maximum effective activity of PRC2, respectively. Here we model all pre-procedures together as a rough Poisson event, whose waiting time follows an exponential distribution, hence the CDF-like form in Equation S4 gives the possibility of successful work of PRC2.

H3K27 demethylation. In plants, there exist several demethylases, e.g., jumonji-C domain-containing (JMJD) protein family, that can catalyze H3K27 demethylation[17,18]. Our model applies non-processive demethylation, with one methyl group removal each time. To simplify the model, we no longer considered JMJD alone in the simulation, instead we used a combined demethylation rate, γ_{dem} as the integrated effect of JMJD-dependent and random removal of methyl groups. Therefore, each histone i undergoes demethylation with propensity

$$r_i^{\mathrm{dem}} = \gamma_{\mathrm{dem}}\left(\delta_{H_i,\mathrm{me1}} + \delta_{H_i,\mathrm{me2}} + \delta_{H_i,\mathrm{me3}}\right) \qquad \text{(Equation S5)}$$

Furthermore, the model assumes that H3K27 demethylation is also coupled with gene expression event in two ways[1,2]. On one hand, every transcription can result in random histones methylation removal (one methyl groups at a time) at each histone as the RNA polymerase moving[19], with probability denoted as P_{dem} per histone. Therefore, the process is shown as

$$\text{with probability } [P_{\mathrm{dem}}]\text{: } H_i \to \begin{cases} \mathrm{me0}, & \text{if } H_i = \mathrm{me1} \\ \mathrm{me1}, & \text{if } H_i = \mathrm{me2} \\ \mathrm{me2}, & \text{if } H_i = \mathrm{me3} \end{cases} \qquad \text{(Equation S6)}$$

On the other hand, histone turnover is positively correlated with gene transcriptional activity[20,21], hence we refreshed each nucleosome (a pair of histones) at each time with exchange probability P_{ex} per histone independently. As a result, the probability that a nucleosome will not be replaced is the product of its two histones, i.e., $(1 - P_{\mathrm{ex}})^2$, therefore the turnover process is shown as

$$\text{with probability } [1-(1-P_{\text{ex}})^2]: \begin{cases} H_{i-1}/H_i \rightarrow \text{me0/me0}, \text{ if } i \text{ is even} \\ H_i/H_{i+1} \rightarrow \text{me0/me0}, \text{ if } i \text{ is odd} \end{cases} \qquad \text{(Equation S7)}$$

STM Expression Cycle

Gene transcription. *STM* expression, as an important component of our model, is closely linked to epigenetic modification. Meanwhile, STM protein itself also plays an important role in maintaining the undifferentiated state of pluripotent cells in shoot apical meristem (SAM) and/or axillary meristem (AM) [22,23]. To investigate the repressive effect of epigenetic modification on gene transcription, we evaluated the dependence of mRNA production on H3K27me3/me2 levels. The transcriptional initiation rate f is a piecewise linear function of the repression level across the *STM* locus, multiplied by another two activation factors (discuss later). Altogether, the formula is given by

$$f = \alpha\theta\left[f_{\max} - \min\left\{\frac{P_{\text{me2/me3}}}{P_{\text{t}}}, 1\right\} \cdot (f_{\max} - f_{\min})\right] \qquad \text{(Equation S8)}$$

where $f_{\max}$ ($f_{\min}$) is the upper (lower) bound of transcription initiation rate, $P_{\text{me2/me3}}$ is the proportion of H3K27me3/me2

$$P_{\text{me2/me3}} = \frac{1}{N_{\text{h}}}\sum_{j=1}^{N_{\text{h}}}\left(\delta_{H_j,\text{me2}} + \delta_{H_j,\text{me3}}\right) \qquad \text{(Equation S9)}$$

The threshold, P_{t} in Equation S8 captures the sub-saturating level for H3K27me3/me2 to reach the maximal repression effect[24], the rationality of which was justified by fitting mass spectrometry data from stable isotope labeling by amino acids in cell culture (SILAC)[1,2,25].

In *Arabidopsis*, a previous study revealed a self-sustained loop of *STM* gene in leaf axil to maintain its expression: STM interacts with ATH1, expressed by gene *ARABIDOPSIS THALIANA HOMEOBOX GENE1* (*ATH1*), to form a ATH1-STM complex, subsequently this complex binds to the *STM* locus by ATH1 and activates gene transcription[26]. This process is reminiscent of a ligand-receptor interactions, hence we defined a self-activation factor for this path, denoted as θ in Equation S8, using the Hill equation

$$\theta = \frac{\theta_{\max} N_{\text{p}}^{\sigma}}{K_d + N_{\text{p}}^{\sigma}} \qquad \text{(Equation S10)}$$

where $\theta_{\max}$, K_d and σ is the maximum self-activation level, apparent disassociation constant and Hill coefficient, respectively. Note that, $\theta_{\max}$ is determined by the affinity between ATH1-STM complex and *STM* locus, we assumed this parameter will quickly saturate within a reasonable improvement, due to the finite length of *STM* gene and steric effect of binding, as the number of ATH1 protein molecule increases in the system. Moreover, we set $\sigma > 1$ to reflects the synergistic effect.

Despite the self-activation, other potential trans-acting regulator of *STM* expression, e.g., *REVOLUTA* (*REV*)[27], *DORNRÖSCHEN* (*DRN*) and *DORNRÖSCHEN LIKE* (*DRNL*)[27,28], Auxin Response Factor (ARFs)[29], etc., can be encoded in this model as extra multipliers α_X (X stands for specific factors) as well. However, since they are inconsequential here, we use a single trans-activation factor $\alpha = \prod_X \alpha_X$ in Equation S8 to represent the integrated effects, where $\alpha < 1$ is repressive, $\alpha = 1$ is neutral, and $\alpha > 1$ is activating, respectively.

Previous model set an upper limit f_{lim} for the transcription initiation rate to restrict it in a reasonable range when $\alpha\theta >> 1$[1,2]; however, we found that this cannot reproduce all of our experimental results. Instead, we successfully addressed our problem by adding a tiny term $\gamma_{\text{transcr}} << 1$ for noisy transcription. Finally, the propensity of *STM* gene transcription is

$$r^{\text{transcr}} = \min\{f, f_{\text{lim}}\} + \gamma_{\text{transcr}} \qquad \text{(Equation S11)}$$

In the end, transcription events are coupled with another two events: demethylation and histone exchange, which we have already elaborated above in the section *H3K27 demethylation*.

mRNA translation. To simplify the model, we no longer considered the conversion of the *STM* gene to mRNA (transcription) and of the mRNA to STM protein (translation) as two split processes, instead we combined them into a single step, i.e., gene expression. The purpose of this approach is to eliminate an extra Gillespie's simulation of intermediate state mRNA[30], which is actually not important in this model mechanism. However, to integrate transcription and translation, we introduced another parameter n_{ppt} to denote the average protein molecule translated per mRNA transcript (where "ppt" stands for protein per transcription). Once a gene transcription event occurred in the simulation, we added the STM protein number by n_{ppt}. Thus, the protein number in our model is more like a generalized concept, as n_{ppt} can also take float values. Here we ignored the time interval between each translation of the same mRNA transcript.

Protein degradation. We assumed the degradation probability was constant in our model. Therefore, the propensity for protein degradation is proportional to the remaining protein molecule number in the system, which is computed as

$$r^{\text{degr}} = \kappa N_{\text{p}} \qquad \text{(Equation S12)}$$

where κ is the degradation coefficient. Once a protein degradation event occurred in the simulation, we subtracted the number of STM protein molecule by 1. If the remaining molecule number is less than 1 before subtraction (possibly happen when n_{ppt} is not an integer), then we set it to 0 to avoid generating a negative value.

DNA Replication and Cell Division

In the simulation, DNA replication is inserted into the Gillespie process as a truncated event when time of the next coming event exceeds the next replication time[31]. Actually, experiments suggested that H3/H4 tetramers did not dissociate and were shared evenly between daughter strand[32]. Therefore, when replication is carried out, each nucleosome (a pair of histones) is updated with a unmethylated one (dilution) with a probability of 0.5[1,2,24]. After each DNA replication, since we only tracked one daughter strand randomly

$$\text{with probability } [0.5]\text{: } H_{2j-1}/H_{2j} \rightarrow \text{me0}/\text{me0} \qquad \text{(Equation S13)}$$

We performed equal-scaling transformation when cell cycle changed in simulation, e.g., if a cell had 1.5 hours to the next division, and then cell cycle doubled, the transformed cell divided 3 hours later. Moreover, previous works suggested that the normal cell cycle T_0 in meristem typically ranged from 18-36 hours[33,34], and we adopted 22.0 hours in this model to follow previous works[1,2].

Additional Details

Gene activity. Defined as the average transcription number in a cell population per 15 minutes interval.

Cell division arrest. To simulate the situation when cells stop dividing by specific inductor, we allowed the cell cycle T to be infinite in our simulations. If cell division is arrested, DNA replication events are excluded from the Gillespie process.

Protein stable/metastable point. According to our model, the propensity for *STM* protein degradation is linear, while the propensity of STM protein production (gene expression) is sigmoid and fluctuates between the two extreme epigenetic state. Consider the general case in which $f < f_{\mathrm{lim}}$ (easily ensured by forcing $\alpha\theta_{\mathrm{max}}f_{\mathrm{max}} \leq f_{\mathrm{lim}}$), then the master equation of average protein number $\langle N_{\mathrm{p}}\rangle$ is given by

$$\frac{\mathrm{d}\langle N_{\mathrm{p}}\rangle}{\mathrm{d}t} = n_{\mathrm{ppt}}r^{\mathrm{transcr}} - r^{\mathrm{degr}} = n_{\mathrm{ppt}}\left(\frac{\alpha F\theta_{\mathrm{max}}N_{\mathrm{p}}^{\sigma}}{K_d + N_{\mathrm{p}}^{\sigma}} + \gamma_{\mathrm{transcr}}\right) - \kappa N_{\mathrm{p}} \qquad \text{(Equation S14)}$$

Here $F = F(P_{\mathrm{me2/me3}}) \in [f_{\mathrm{min}}, f_{\mathrm{max}}]$ decreases as $P_{\mathrm{me2/me3}}$ increases. In a steady state when $P_{\mathrm{me2/me3}}$ remains almost constant, the roots of above equation give the fixation points of current system. Regardless, there should be one fixation point (close to 0) due to tiny noisy transcription, which stands for the stably differentiated state (Fig. S1B). However, the other two fixation points (for small $P_{\mathrm{me2/me3}}$) have very different properties. For the larger one, the system reaches a stable equilibrium and tends to recover itself like a spring after perturbation in any direction; thus, we termed this a stable point (Fig. S1A). Actually, this is the case for normal pluripotent cells with stable *STM* expression (Fig. S2B). For the smaller one, a positive disturbance leads to continuous increasing of protein (accompanied by demethylation) until meet the higher stable point, which represents the direction of pluripotency restoration; while a negative disturbance causes irreversible decreasing of protein (accompanied by methylation) until vanish at the lower stable point, which represents the direction of differentiation. Altogether, the equilibrium here is unstable and we named it as metastable/critical point (Fig. S1A). As $P_{\mathrm{me2/me3}}$ rises, the stable point drops while the metastable point elevates until they meet at a branch point and annihilate, indicating that the high-STM (stem cell) homeostasis only exists in low methylation region (Fig. S2B).

Differentiation/pluripotency criteria. We used the remaining number of STM protein molecule in the system to characterize the cell fate. In our simulation, the cells with STM protein left less than 1.0 were viewed as completely differentiated, while the cells with STM protein left more than half of the high stable point of the lowest epigenetic repression (H3K27me0) state were viewed as stem cells, others are considered in a transitional state.

Bistability measurement. Bistability describes the ability to maintain both stem cell fate with low histone repression (active transcription) and differentiated cell fate with high histone repression (inactivate transcription) state of *STM* gene. Here we followed the conception of previous models in which the bistability measurement B[1,2,35], was given by

$$B = 4P_{\mathrm{ON}}P_{\mathrm{OFF}} \qquad \text{(Equation S15)}$$

where P_{ON} stands for the probability of the *STM* gene being in the epigenetic ON state, which is defined as

$$P_{\mathrm{ON}} = \Pr\left(P_{\mathrm{me2/me3}} < \frac{P_{\mathrm{t}}}{4}\right) \quad \text{(Equation S16)}$$

and likewise, P_{OFF} represents the probability of the *STM* gene being in the epigenetic OFF state

$$P_{\mathrm{OFF}} = \Pr\left(P_{\mathrm{me2/me3}} > \frac{3P_{\mathrm{t}}}{4}\right) \quad \text{(Equation S17)}$$

In practice, B is calculated using Equation S15 from the combined outputs from the equal number of simulations initiated from either stem cells (uniform H3K27me0, STM protein at high stable point level) or differentiated cell (uniform H3K27me3, no STM protein). B values closer to 1 indicate stronger bistability. For reliability, we allowed the system to recover from the perturbation of DNA replication before sampling, thus only the data from steady state (last hour of each cell cycle) was used in computation[1,2].

Cycle-related parameter. Set at least one dynamics parameter with cell cycle-dependency is necessary in tests, which can provide a source force to switch from one of the two stable state of a bistable model to another when the cell cycle is altered. In our implementation, we picked the effective activity of PRC2 as the correlated parameter $\beta = \beta(T)$, for it has a reasonable explanation and simple form, as we discussed in the main text.

As a contrast, we considered an alternative model using a constant $\beta = \beta(T_0)$. Simulation results showed in this case the model could not reproduce the cell type transformation during division arrest, for almost all cells still tend to retain the last homeostasis before quiescence, i.e., the low-methylation state (Fig. S7A-D).

Programming Languages and Code Resources

The model programs were written in Python 3.13. All simulations were tested both on the Windows (win11) and Linux (Ubuntu 24.04 LTS) operating system. The source code of the model is available at https://github.com/yy420106/STMHybridModel.

Table S1. Summary of model parameters

Parameter	Biological Description	Used Value	Notes/References
Global			
N_h	Number of H3 histones	36	Fixed
N_p	Number of STM protein molecules	≥ 0	
Cell Division			
T	Cell cycle (inverse of cell division frequency) [hour]	> 0	
T_0	Normalized cell cycle [hour]	22.0	Berry et al.[1]; Lövkvist et al.[2]; Grandjean et al.[33]; Reddy et al.[34]
Gene activation			
α	Trans-acting gene activation of *STM*	1.0	Berry et al.[1]; Lövkvist et al.[2]
θ_{max}	Maximum self-activation level of *STM*	1.0	
σ	Hill coefficient	2.0	
K_d	Apparent disassociation constant of ATH1-STM complex	180.0	
Gene transcription & translation			
f_{min}	Minimum transcription initiation rate [1/sec]	10^{-4}	Berry et al.[1]; Lövkvist et al.[2]
f_{max}	Maximum transcription initiation rate [1/sec]	5×10^{-3}	
f_{lim}	Upper bound of transcription initiation rate [1/sec]	$1/60$	
P_t	Threshold proportion of H3K27me3/me2 marks for maximal repression	$1/3$	Berry et al.[1]; Lövkvist et al.[2]; Jadhav et al.[24]; Alabert et al.[25]
$\gamma_{\mathrm{transcr}}$	Noisy transcription rate [1/sec]	1×10^{-8}	
n_{ppt}	Average proteins translated per mRNA transcript	1.0	
Protein degradation			
κ	Protein degradation rate [1/sec]	5×10^{-6}	
Histone methylation			
β_{max}	Maximum effective activity of PRC2	2.5	
b	Normalized exponential factor	$\ln(5/3)$	
e_{distal}	Distal gain by DNA looping	0.001	
ρ	Reduced efficiency of activated PRC2 between H3K27me2/me3	$1/10$	Berry et al.[1]; Lövkvist et al.[2]; Margueron et al.[8]
k_{me}	Reference PRC2-mediated methylation rate [1/sec]	1×10^{-5}	Berry et al.[1]; McCabe et al.[12]

k_{me23}	PRC2-mediated methylation rate (me2 to me3) [1/sec]	k_{me}	
k_{me12}	PRC2-mediated methylation rate (me1 to me2) [1/sec]	$6k_{\mathrm{me}}$	
k_{me01}	PRC2-mediated methylation rate (me0 to me1) [1/sec]	$9k_{\mathrm{me}}$	
γ_{me23}	Noisy methylation rate (me2 to me3) [1/sec]	$k_{\mathrm{me23}}/20$	Berry et al.[1] Lövkvist et al.[2]
γ_{me12}	Noisy methylation rate (me1 to me2) [1/sec]	$k_{\mathrm{me12}}/20$	
γ_{me01}	Noisy methylation rate (me0 to me1) [1/sec]	$k_{\mathrm{me01}}/20$	
Histone demethylation & exchange			
P_{dem}	Transcription-coupled demethylation probability per histone	6.4×10^{-3}	
P_{ex}	Transcription-coupled exchange probability per histone	1.5×10^{-3}	
γ_{dem}	Integrated demethylation rate [1/sec]	$f_{\mathrm{min}}P_{\mathrm{dem}}$	Berry et al.[1]; Lövkvist et al.[2]

Supplementary References

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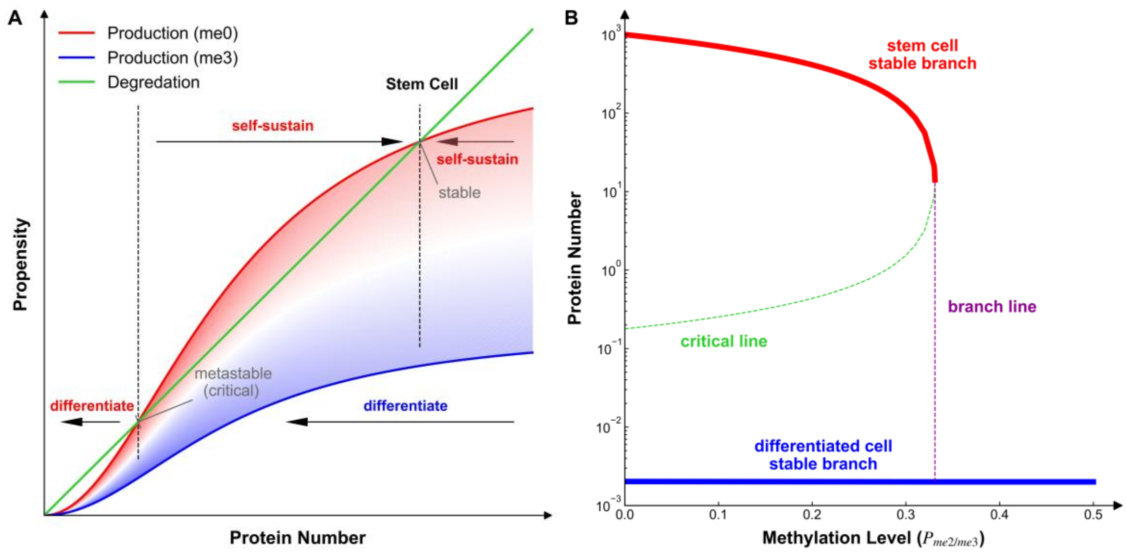


Fig. S1 | Mathematical principles of the model.

(A) Schematic plot explaining the STM protein dynamics. The shaded area represents the variation range of the STM expression curves between two extreme epigenetic states, with a redder/bluer curve representing a lower/higher repression level. The cell reaches a temporally homeostasis when the production curve (sigmoidal) and degradation curve (linear) of STM equals. The minimum (close to 0) and maximum fixation point, which is self-sustained, corresponding to the differentiation and pluripotency state, respectively. The middle fixation point is sensitive to the disturbance, result in different evolution direction, representing the critical state. Note that the graph was not drawn to scale.

(B) Phase diagram explaining the bifurcation of cell fate. Solid blue line represents the stable branch of fully differentiated cells. Solid red line represents the stable branch of stem cells, that annihilates at the left limit of branch line (purple dashed line). Green dashed line represents the trajectory of metastable/critical point, which divide the left space into pluripotency bias and differentiation bias region. On the right side of branch line, where the methylation level is high enough, only differentiated cell stable branch exists.

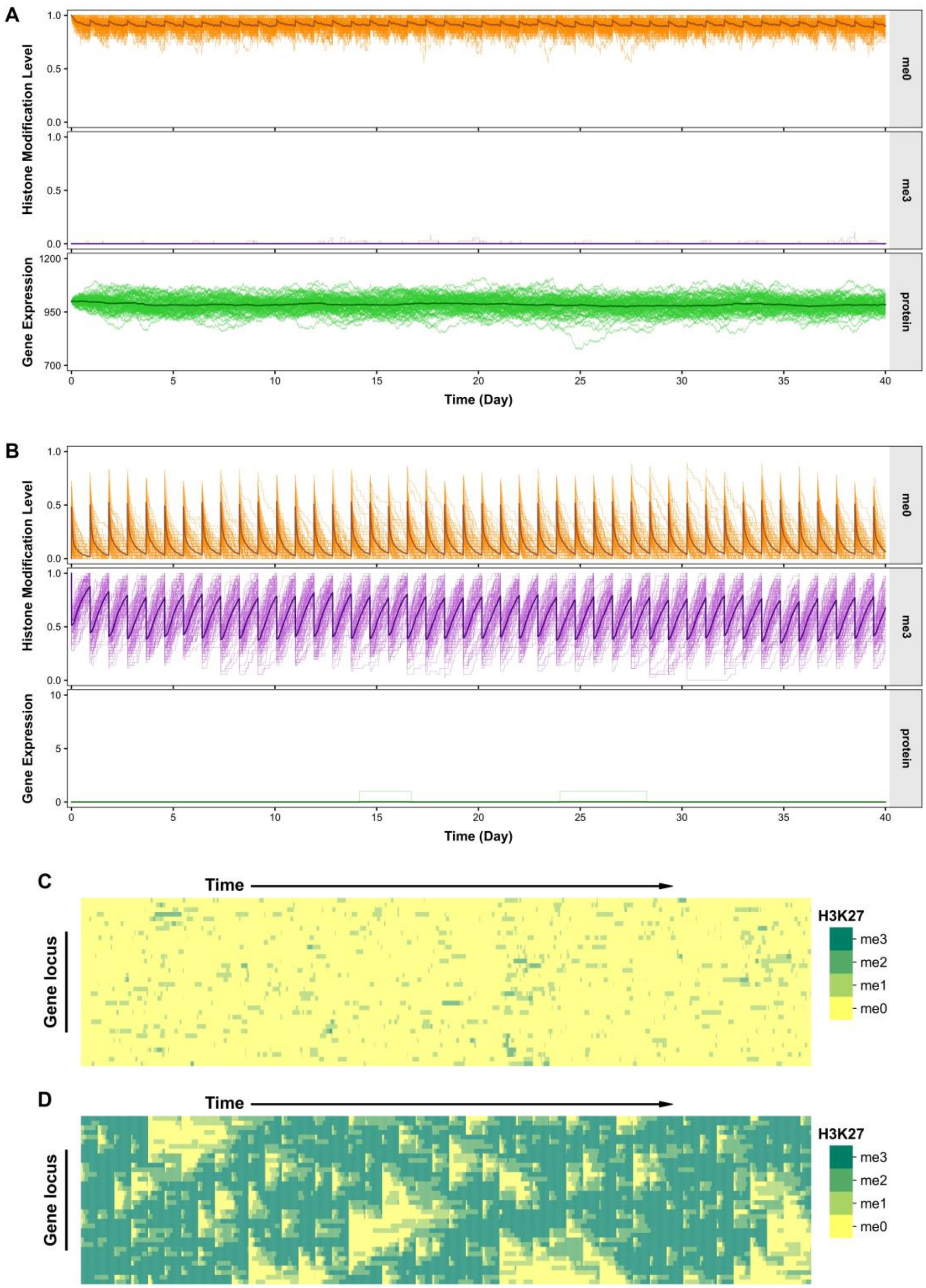


Fig. S2 | Verify the model stability of both pluripotent and differentiated cells.
(A-B) Representative trajectories from stochastic simulations of (A) pluripotent cells (uniform H3K27me0, STM protein at high stable level) and (B) differentiated cells (uniform H3K27me3,

no STM protein), with a normal cell cycle (22 h). Each graph shows 80 overplotted trajectories. Thick lines represent the average curve of the cell population.

(C-D) Representative evolution of methylation marks at *STM* locus from stochastic simulations of a (C) pluripotent cell and (D) differentiated cell. The single cell data were randomly selected from trajectories in (A) and/or (B).

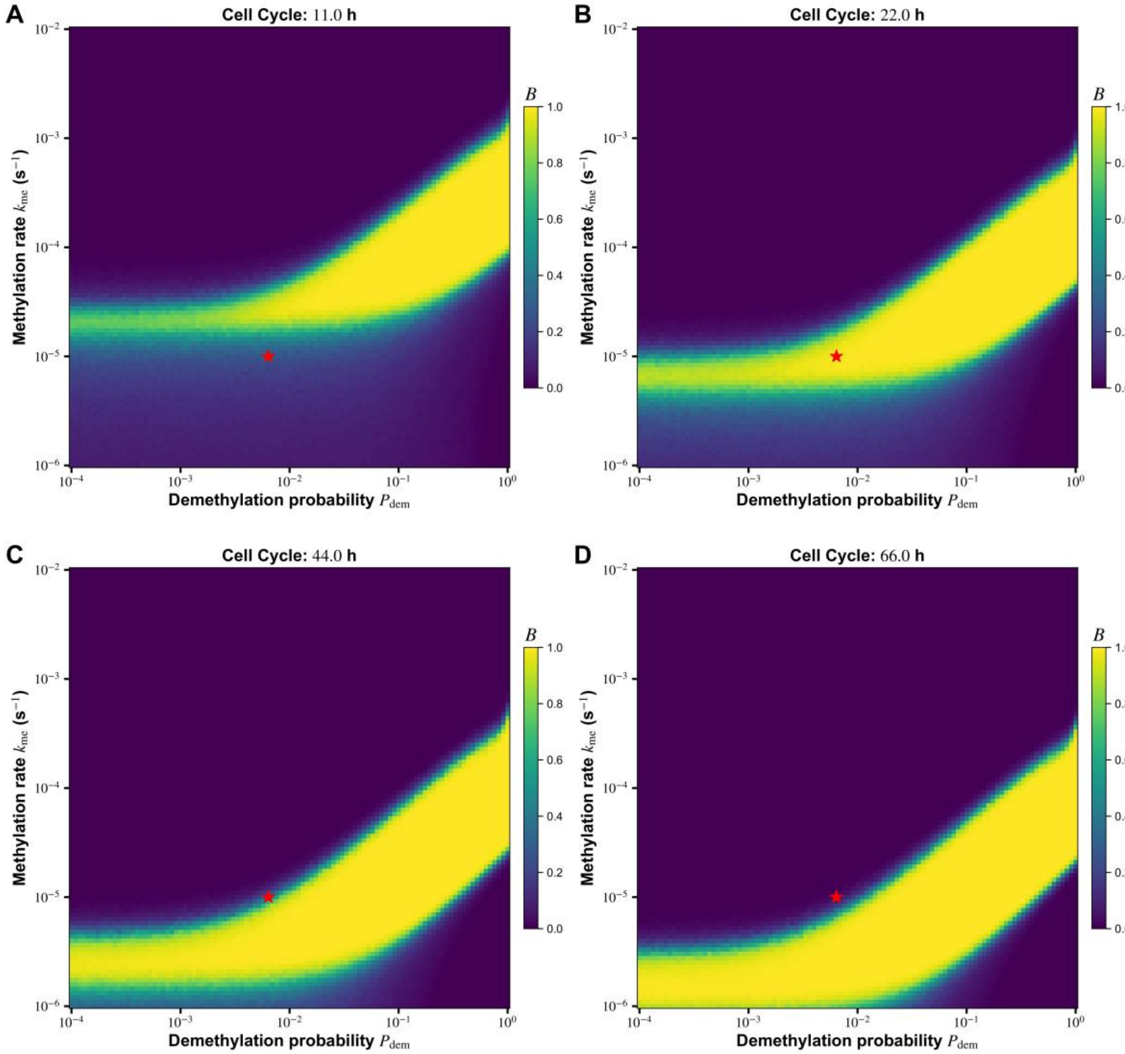


Fig. S3 | Heatmap of bistability distribution against varying cell cycle.

(A-D) Heatmap showing values for the bistability measurement B, calculated from simulations. Each graph shows B as a function of the methylation rate k_{me} and demethylation probability P_{dem}, when the cell cycle is (A) 11 h, (B) 22 h, (C) 44 h and (D) 66 h. B is close to 1 for bistable systems. Red asterisks mark the default values used in the model. For each parameter set, 100 simulations were initialized from either pluripotent cells (uniform H3K27me0, STM protein at high stable level) or differentiated cells (uniform H3K27me3, no STM protein), and simulated for 40 cell cycles. Data within the last hour of each cell cycle was collected for computation, and the pixel values were averaged over all simulations.

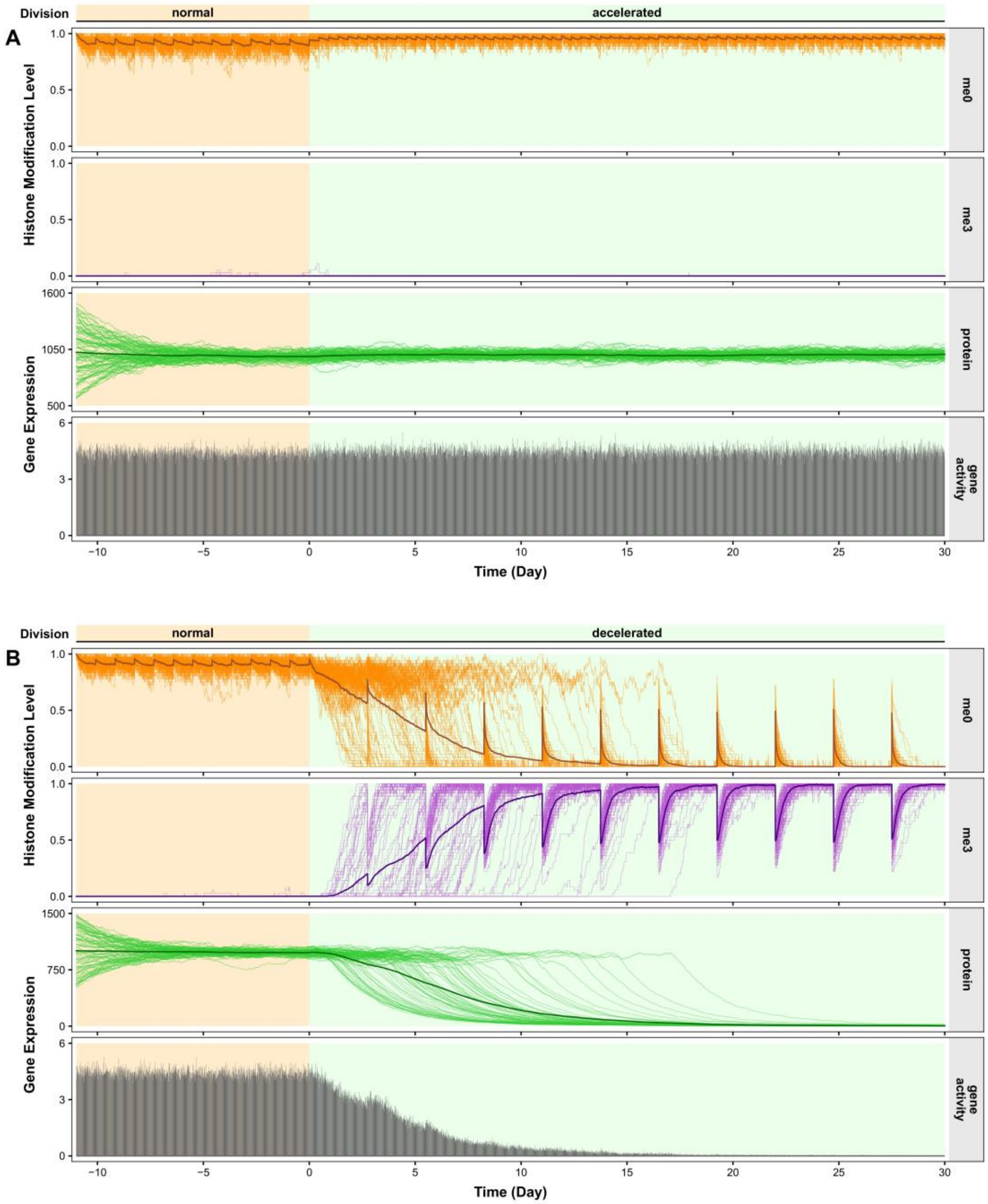
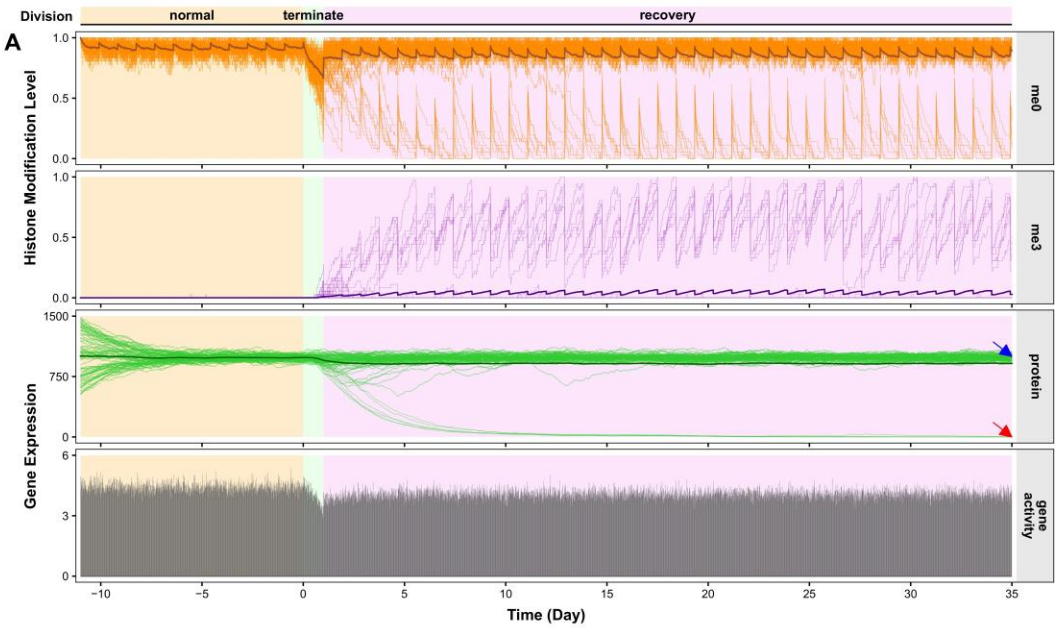
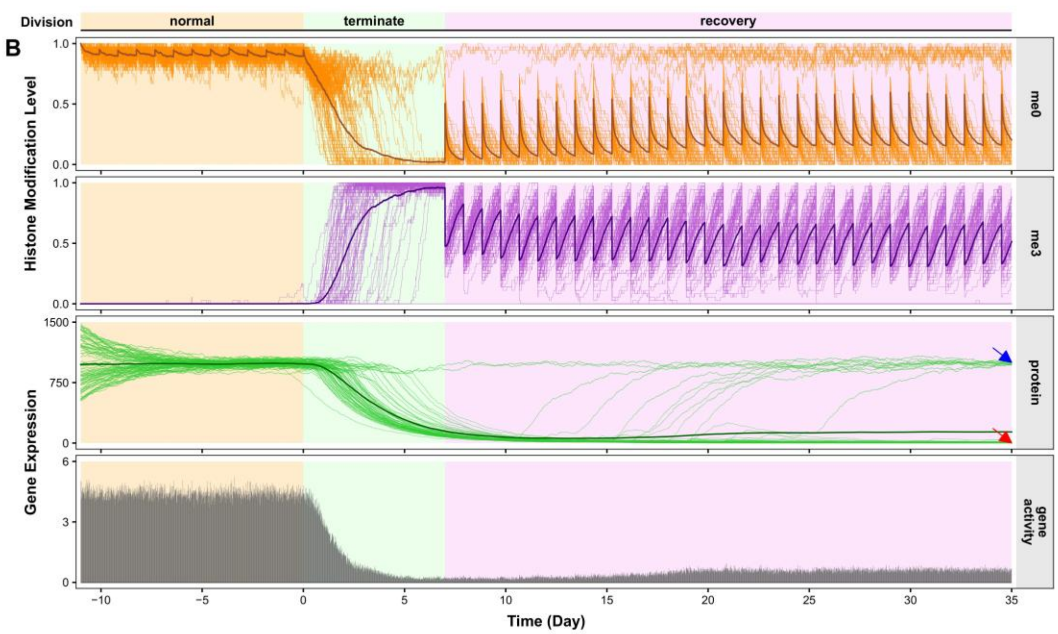


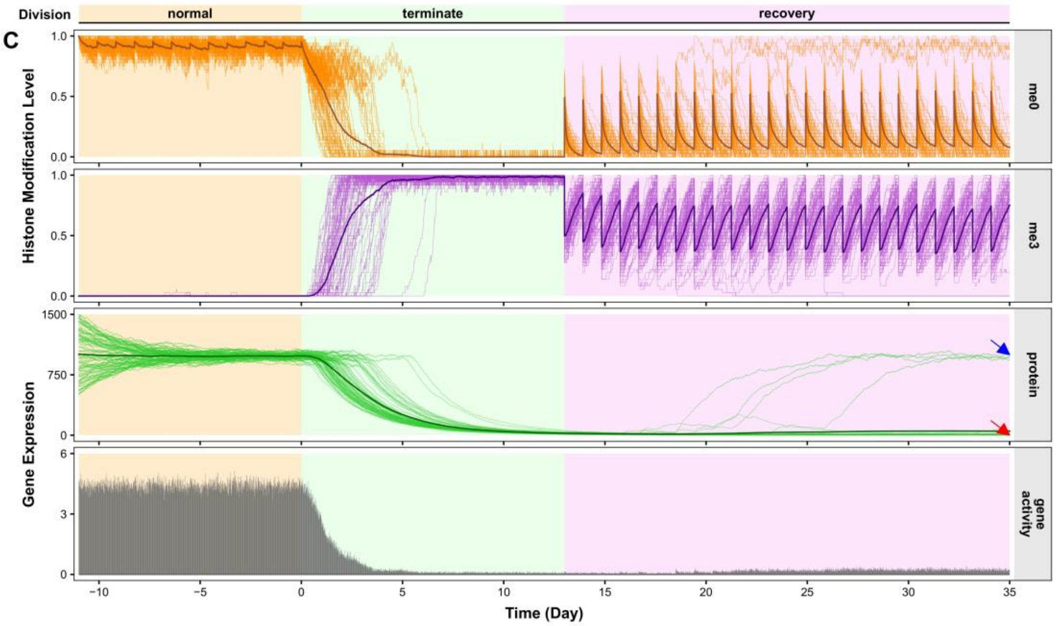
Fig. S4 | Altering cell cycle changes the homeostasis and cell fate.

(A-B) Representative trajectories from stochastic simulations of (A) 2-fold accelerated and (B) 1/3-fold decelerated cell division. In (A), the protein level increased by ~2% (from 981.0±38.1 to 1000.8±32.0). In (B), the protein level decreased by ~99.7% (from 979.6±37.8 to 3.1±5.2). Simulations were started randomly from ±50% fluctuation of protein stable points in pluripotent cells (uniform H3K27me0) with a normal cell cycle (22 h), and 12 cell cycles were performed for pre-equilibration before changing the cell cycle. Each graph shows 80 overplotted trajectories. Thick lines represent the average curve of the cell population.

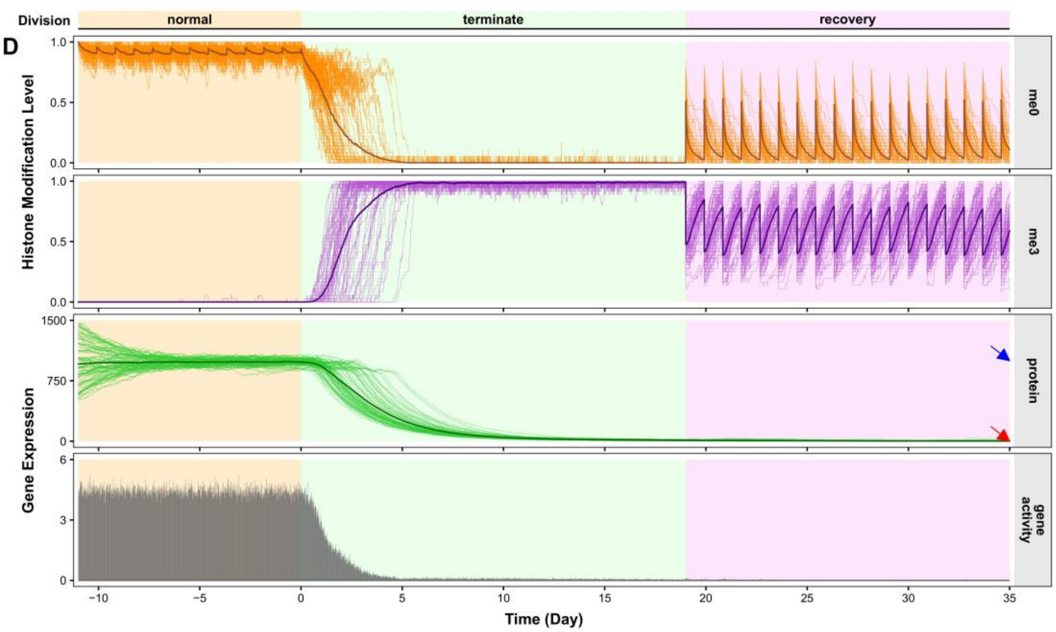

Division
normal
terminate
recovery
A
Histone Modification Level
Gene Expression
me0
me3
protein
gene activity
Time (Day)


Division
normal
terminate
recovery
B
Histone Modification Level
Gene Expression
me0
me3
protein
gene activity
Time (Day)

C


Division
normal
terminate
recovery
Histone Modification Level
Gene Expression
me0
me3
protein
gene activity
Time (Day)

D


Division
normal
terminate
recovery
Histone Modification Level
Gene Expression
me0
me3
protein
gene activity
Time (Day)

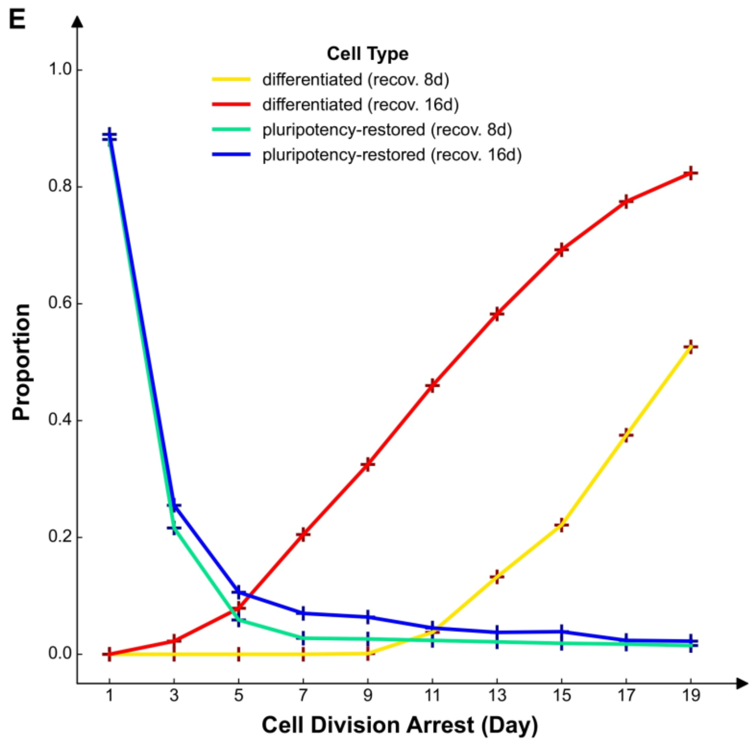
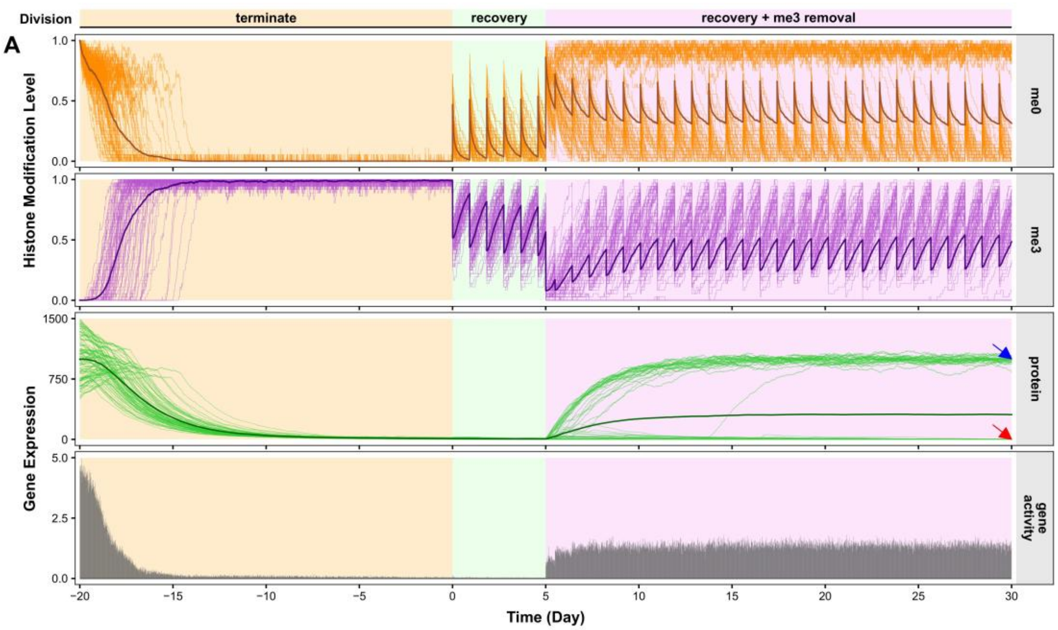


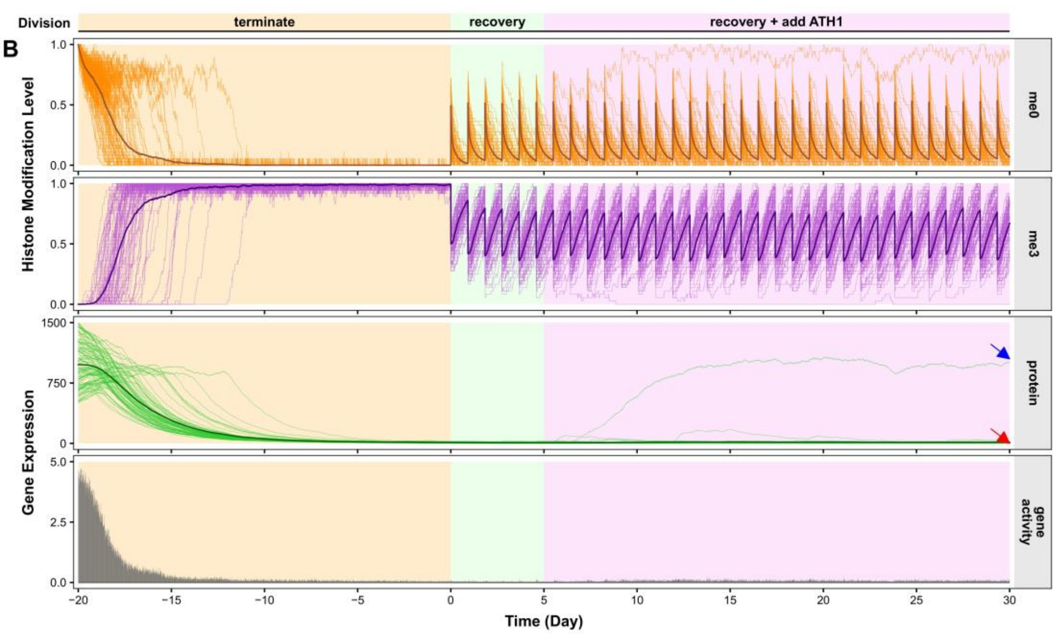
Fig. S5 | Prolonging division arrest leads to irreversible differentiation of stem cells and weakens their capacity to restore original pluripotency.

(A-D) Representative trajectories from stochastic simulations of (A) 1 d, (B) 7 d, (C) 13 d and (D) 19 d of cell division arrest before recovery. Simulations were started randomly from ±50% fluctuation of protein stable points in pluripotent cells (uniform H3K27me0) with a normal cell cycle (22 h), and 12 cell cycles were performed for pre-equilibration before quiescence. Each graph shows 80 overplotted trajectories. Thick lines represent the average curve of the cell population. Red arrowheads mark differentiated group; blue arrowheads mark pluripotency-restored group.

(E) Curves showing the proportion of differentiated/pluripotency-restored cells on the 8th and/or 16th day after the completion of different lengths of division arrest. Abbreviation [recov. Xd] stands for X d of redivide after quiescence. For each arrest length, the point values were averaged over 800 simulations. Some trajectories were plotted in (A-D).



Division
terminate
recovery
recovery + me3 removal
A
Histone Modification Level
me0
me3
Gene Expression
protein
gene activity
Time (Day)



Division
terminate
recovery
recovery + add ATH1
B
Histone Modification Level
me0
me3
Gene Expression
protein
gene activity
Time (Day)

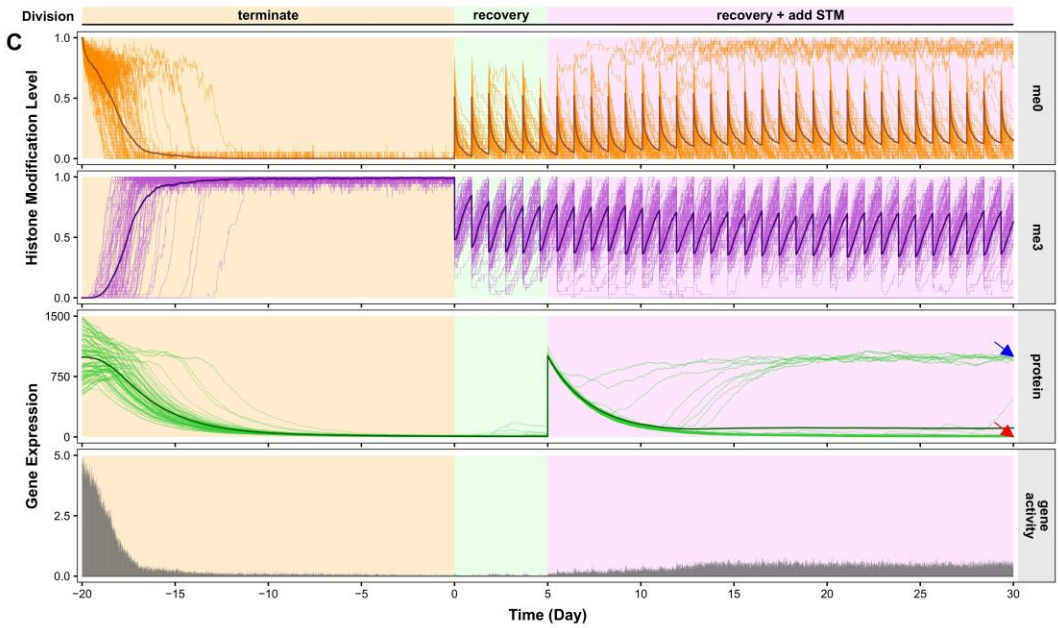
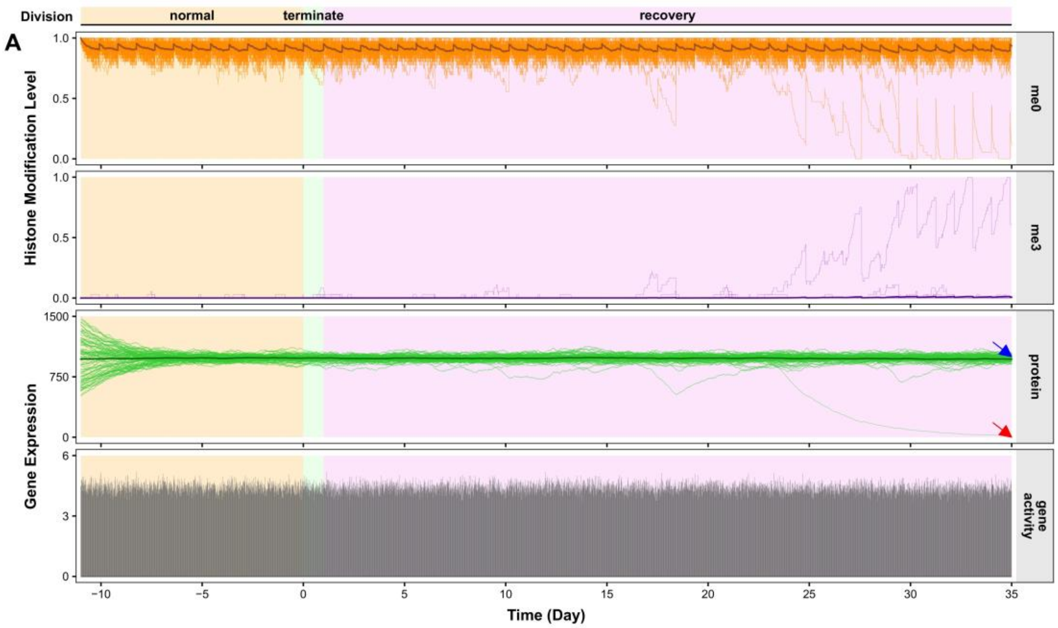
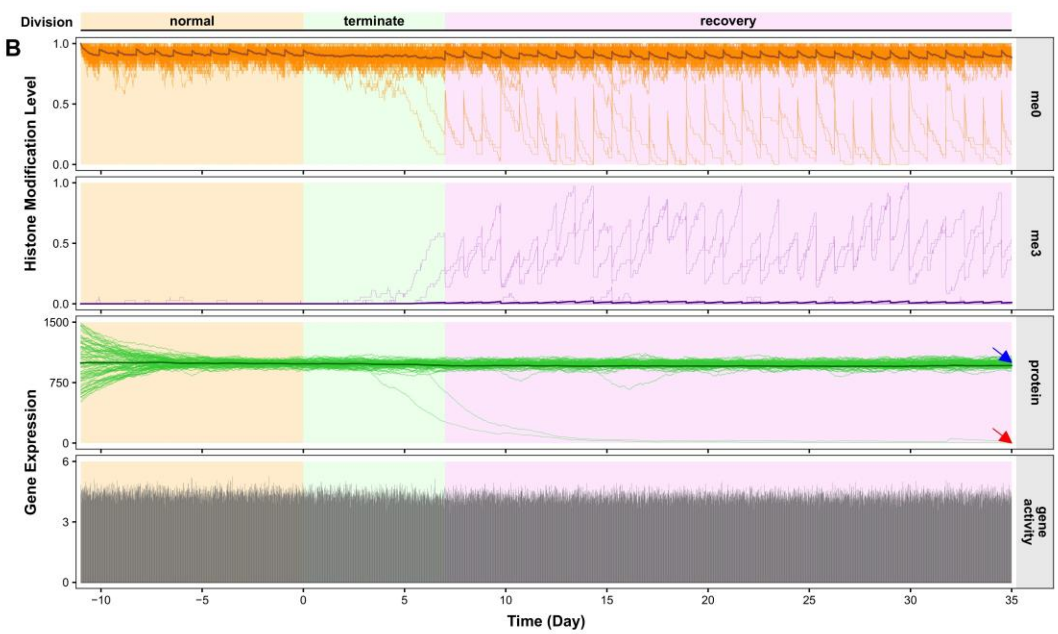


Fig. S6 | Model prediction of different rescue strategies of differentiated cells.

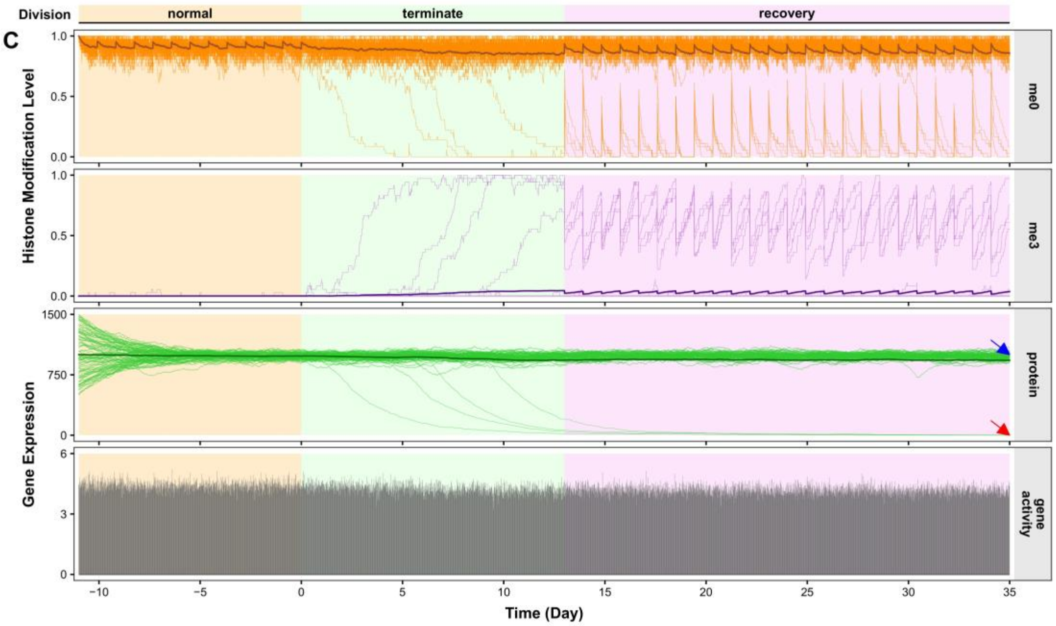
(A-C) Representative trajectories from stochastic simulations of (A) H3K27me3 removal (reset each histone to me0 with a probability of 85%), (B) *ATH1* overexpression (increased the maximum self-activation level of STM, i.e., $\theta_{\max}$, by 5%) and (C) exogenous injection of STM mRNA/protein (returned the protein to normal level) after long-term quiescence. Simulations were started randomly from ±50% fluctuation of protein stable points in pluripotent cells (uniform H3K27me0), after which division was stopped for 20 d, and the cells were then allowed to rest for another 5 d before rescue treatment. Each graph shows 80 overplotted trajectories, with (A) 25 samples, (B) 1 sample and (C) 8 samples regained pluripotency. Thick lines represent the average curve of the cell population. Red arrowheads mark differentiated group; blue arrowheads mark pluripotency-restored group.



Division
normal
terminate
recovery
A
Histone Modification Level
me0
me3
Gene Expression
protein
gene activity
Time (Day)



Division
normal
terminate
recovery
B
Histone Modification Level
me0
me3
Gene Expression
protein
gene activity
Time (Day)



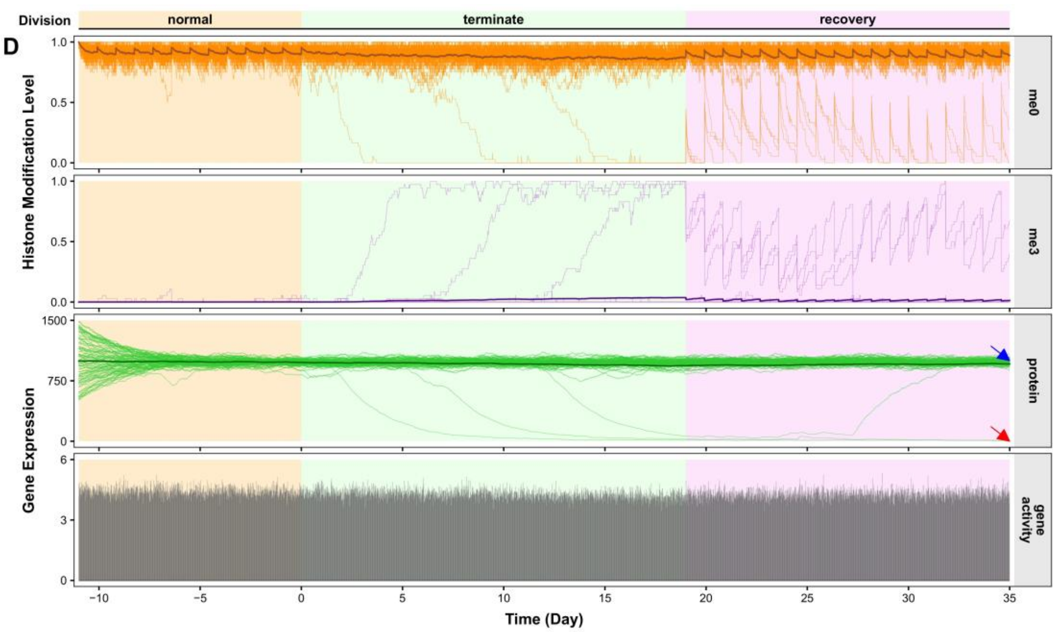


Fig. S7 | Alternative model without cycle-related parameters cannot reproduce the cell type transformation during division arrest.

(A-D) Representative trajectories from stochastic simulations of (A) 1 d, (B) 7 d, (C) 13 d and (D) 19 d cell division arrest before recovery using the alternative model, where the effective activity of PRC2, i.e., β, is fixed to a constant $\beta(T_0)$ (Equation S4). Other parameters are the same as before. Simulations were started randomly from ±50% fluctuation of protein stable points in pluripotent cell (uniform H3K27me0) with a normal cell cycle (22 h), and 12 cell cycles were performed for pre-equilibration before quiescence. Each graph shows 80 overplotted trajectories. Thick lines represent the average curve of the cell population. Red arrowheads mark differentiated group; blue arrowheads mark pluripotency-restored group.